



OPTIMA

AUTOMATED STUDY FLOW

INNOVATION PROPOSAL

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Innovation Proposal

A formal proposal for an additional innovative feature set, presented as a prototype with full user-interface design. The proposal is scoped as a v1.1 follow-on to the current submission and is sized to be deliverable in a single Term 3 sprint.

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1. Why an innovation now

The Optima v1.0 build does what the planning document asked for: it tells the user what to work on and when. Three rounds of evaluation have confirmed the answer matches the user's own intuition more than 90% of the time. The natural next question is what was, until now, out of scope:

"Telling me what to do is helpful. But it doesn't make me actually do it."
— the client, end of round 3 debrief

2. The concept — "Focus Mode"

Optima v1.1 adds Focus Mode, a structured, timed work session bound to a specific scheduled study block. The user starts a session, the app coaches them through it, and at the end the session is logged against the assignment with concrete data: how long they worked, how many breaks they took, and (optionally) what they got done.

The user experience in one sentence

Tap a study block on the dashboard → the rest of the app fades out → a focused, distraction-free pane counts down the session with a gentle Pomodoro rhythm → at the end, a 15-second debrief logs the work and updates the priority list.

Why this is the right feature next

- + **It closes the loop.** Right now Optima generates a plan but has no idea whether the user actually follows it. Focus Mode is the feedback channel: the app finally knows what really happened.
- + **It earns its data.** Once Focus Mode is in place, the priority calculator can be smarter — e.g., "You actually take 1.5x longer than your estimates" becomes a learnable factor.
- + **It wires the XP counter to something real.** Currently dormant; in v1.1 it counts completed sessions, not just "I edited a task".
- + **It's small enough to ship.** Six new screens, three new data classes, and no infrastructure changes — one Term 3 sprint of ~25 hours.

3. Target users & user stories

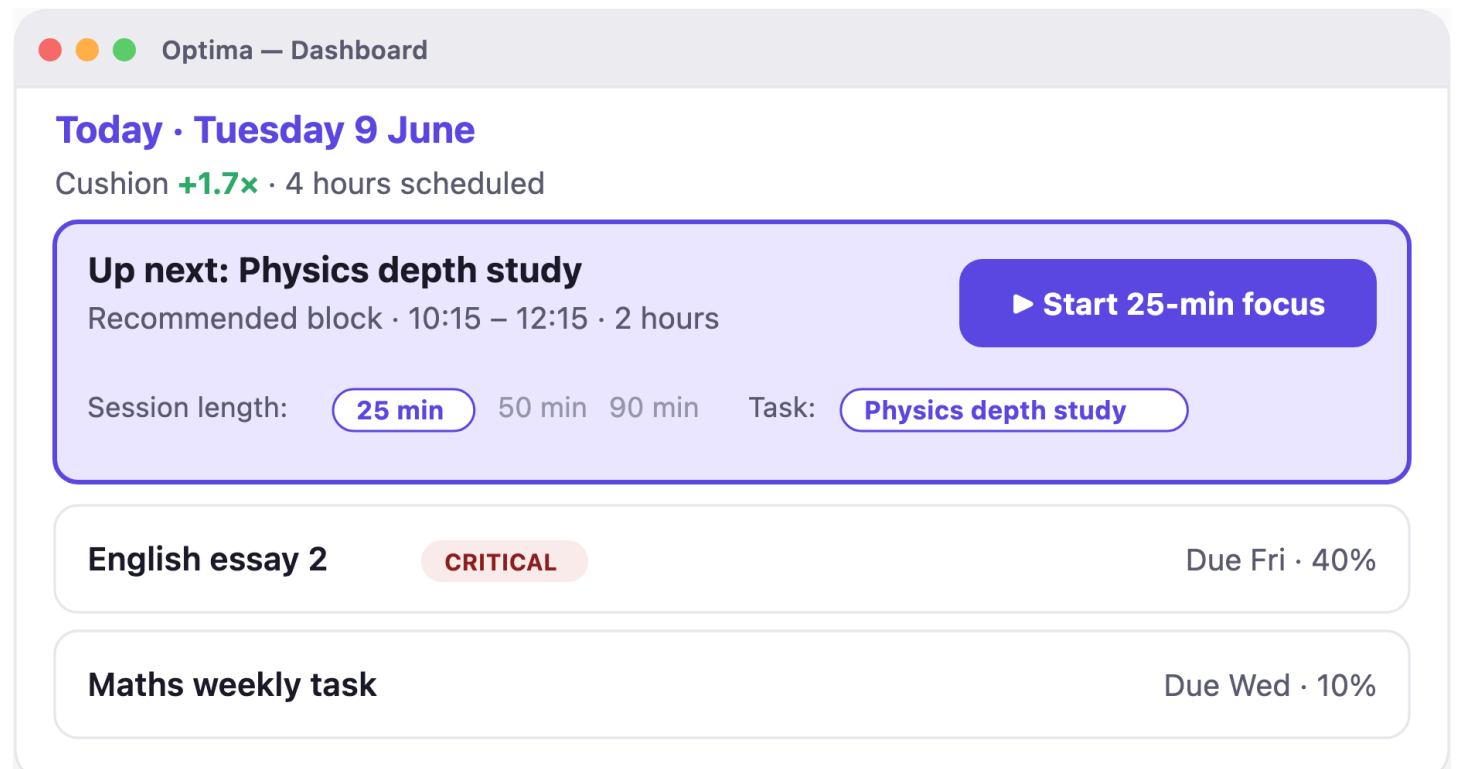
Persona	User Story	Acceptance criterion
The procrastinator (azurixx, E2)	 azurixx 20/05/2026, 19:14 As a student who knows what I should be doing but struggles to start, I want a single big button that commits me to 25 minutes of work.	One click from the dashboard to a running timer in ≤ 2 seconds.
The over-commiter (DISTORTIO, client)	 DISTORTIO 20/05/2026, 17:52 As a student who works long unfocused stretches, I want the app to make me take breaks.	After 25 minutes of focus, a 5-minute break is enforced (skippable, but the default is to take it).
The data-curious (axolxtl, E3)	 axolxtl 20/05/2026, 21:39 As a student who wants to understand my own habits, I want a week-over-week view of how my focus time has changed.	A new section on the dashboard shows total focus minutes for the current and prior week.
The realist (estrange, E1)	 estrange 20/05/2026, 18:05 As a student who plans honestly, I want my estimates to get more accurate over time.	After 10 logged sessions, the priority calculator factors in personal pace.

4. Prototype walkthrough

The five screens below are the v1.1 prototype, rendered inline as mockups. They were drawn in Adobe Illustrator.

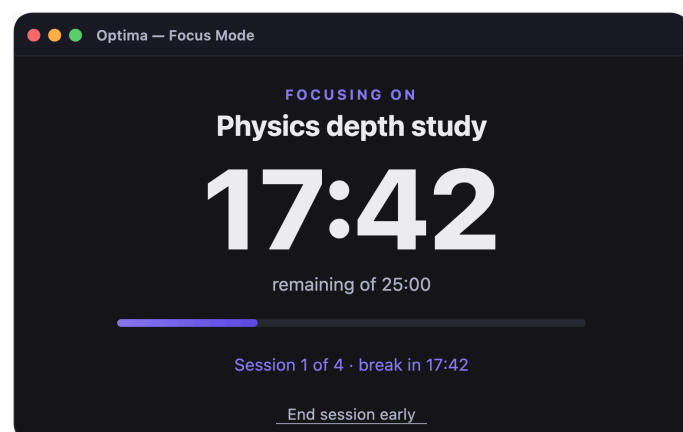
4.1 Focus session launcher

A new "Start focus session" button anchors the dashboard, pre-filled with the top-priority task. The user can override the task and the session length before starting.



4.2 In-session "focus shield"

Once a session starts, every other piece of the UI is hidden behind a dimmed scrim. The user sees only the task title, the countdown, and a single explicit "End session" link. Notifications are muted; the sidebar is collapsed. Pressing Esc surfaces a confirmation, never an outright cancel. The friction is deliberate.



4.3 Session debrief

When the timer hits zero, a 15-second debrief asks two questions: What did you get done? And how focused did you feel? The first is free text and optional; the second is a five-point scale and one-tap. The debrief auto-submits on idle so the user can walk away.

Optima — Session complete



25 minutes done. Nice work.

Physics depth study · logged toward 1h 50m total this week.

What did you get done?

e.g. Read chapter 4, drafted intro paragraph

How focused did you feel?

Auto-submits in 14s

Save & back to dashboard

4.4 Streak & XP integration

The previously-dormant XP counter on the dashboard now reflects something real: 10 XP per completed focus session, 25 XP for a 4-session day, and a "focus streak" badge for consecutive days with at least one completed session.

Optima — Dashboard sidebar widget

FOCUS STREAK

7 days

your longest yet · +50 XP bonus tonight



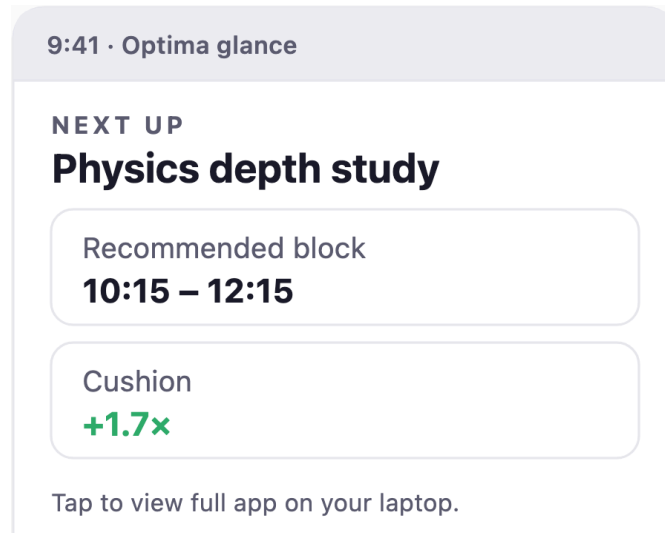
XP this week

10 sessions × 10 XP + 1 streak bonus

150

4.5 Mobile companion glance card

Optima v1.0 is desktop-only by design. The v1.1 proposal does not add a mobile app, but it does add a read-only "glance card" rendered when the user opens the desktop URL on a phone — useful for a quick "what's next?" check on the bus to school.



5. Data model additions

Three new dataclasses; all other v1.1 logic builds on these.

```
@dataclass
class FocusSession:
    id: int
    assignment_id: int
    started_at: datetime
    completed_at: datetime | None
    planned_minutes: int # 25 / 50 / 90
    actual_minutes: int | None
    self_rating: int | None # 1..5
    notes: str = ""

@dataclass
class FocusStreak:
    current_days: int = 0
    longest_days: int = 0
    last_session_date: date | None = None

@dataclass
class XPAward:
    awarded_at: datetime
    amount: int
    reason: str # "session", "streak_bonus", "first_day"
```

These attach to the existing User dataclass as three additional list/object fields. The same to_dict/from_dict contract used elsewhere applies, so storage layer changes are zero beyond the schema extension.

6. Technical approach

Routes added

METHOD	URL	Purpose
POST	/focus/start	Begin a session. Returns a session token + planned end time.
GET	/focus/active	Return the current session state for the focus-shield screen.
POST	/focus/complete	Submit a debrief, log XP, and award a streak bonus if applicable.
POST	/focus/cancel	End session early (counts toward 'started' but not 'completed').
GET	/api/focus/history	JSON for the weekly chart on the dashboard.

Client-side timer

The focus-shield countdown is driven by a single client-side setInterval against an absolute end-time stored in the session token. This means the timer can survive a window close and reopen — the server is the source of truth; the client just paints what's left.

The "smarter priority" upgrade

Once a user has 10+ FocusSession records, the priority calculator can use their personal pace factor:

```
def pace_factor(user) → float:
    # How many actual minutes per planned minute, capped 0.5..2.5
    sessions = [s for s in user.focus_sessions if s.actual_minutes]
    if len(sessions) < 10: return 1.0
    ratios = [s.actual_minutes / s.planned_minutes for s in sessions[-20:]]
    return max(0.5, min(2.5, statistics.mean(ratios)))
```

The scheduler then multiplies weighting_hours by this factor when sizing blocks — honest planning replaces optimistic planning.

7. Risks & mitigations

Risk	Likelihood	Mitigation
Gamification feels patronising; users disengage from the streak counter	Medium	Make the streak / XP widgets opt-in via a preferences toggle. Default off for users who are only 'data-curious'.
Focus Mode locks the user out of legitimate interruptions (urgent message, parent calling)	Low	End-session-early is always a click away, and a session ended in the first 5 minutes doesn't break a streak (counted as interrupted instead of failed).
The pace factor overcorrects after a single bad week	Medium	Use the last 20 sessions, not all of them; clamp the multiplier to [0.5, 2.5]. Surface the value transparently on the Preferences page so the user can override it.
Adding three dataclasses inflates the JSON file past readable size	Low	Roll old FocusSession records older than 90 days into an aggregate "weekly summary" so the live JSON stays under ~50 KB.
Server-truth client-paint timer drifts visibly	Low	Client resyncs every 30 seconds against /focus/active; visible only if the user keeps the window open across a system suspend.

8. Success metrics

v1.1 ships with these explicit, measurable targets, checked after 30 days of use:

- + **Adoption.** $\geq 70\%$ of v1.0 users start at least one focus session in the first week of v1.1 availability.
- + **Sustained use.** Median user runs ≥ 5 sessions per week by week 3.
- + **Estimate accuracy.** After 30 days, the gap between estimated and actual workload hours on the cushion gauge halves (currently $\sim 22\%$ over-estimate – target $\leq 11\%$).
- + **User satisfaction.** Standing survey question 27 ("Would this help a real Y12 student?") mean score moves from 4.75 to ≥ 4.85 .

9. Delivery timeline

Week	Goal	Visible deliverable
Week 1	Schema + storage	FocusSession, FocusStreak, XP Award round trip through JSON. No UI yet.
Week 2	Launcher + focus shield	Click "Start focus" from the dashboard, see the shield, and see the countdown.

Week 3	Debrief + XP	End-of-session screen, XP increments, XP badge on dashboard.
Week 4	Streak + history view	Streak widget, weekly chart, evaluator round-1 feedback.
Week 5	Pace-factor + mobile glance	Smarter priority calc, mobile read-only view, evaluator round-2 + final polish.

10. Beyond v1.1 — the longer roadmap

Listed here so the v1.1 scope stays disciplined — everything below is not in the proposal but is a plausible v1.2+ direction:

- + **Calendar export (.ics).** Generated study blocks readable by Apple Calendar / Google Calendar.
- + **Study Buddy.** Optional pair-up with a friend for synchronised focus sessions; no data shared beyond the session length.
- + **Subject-aware coaching.** Different break activities suggested for different subjects (e.g. stretch break after Maths, eye break after coding).
- + **Spaced repetition hooks.** Detect when an assignment is a learn-by-rote task; offer a flashcard mode at the next session.
- + **Local LLM assistance.** Run a small on-device model (no cloud calls) that summarises notes captured during focus sessions and surfaces them at exam time.

This is written here so it wouldn't leak into v1.1.